

Statistics 6910: Representation of Information

Representation of Numbers

Bits and Characters

Binary Representations of Characters				
Glyph	ASCII	UniCode	Glyph	Unicode
#	0010 0011	0000 0000 0010 0011	©	0000 0000 1010 1001
\$	0010 0100	0000 0000 0010 0100	æ	0000 0000 1110 0110
A	0100 0001	0000 0000 0100 0001	Δ	0000 0011 1001 0100
a	0110 0001	0000 0000 0110 0001	α	0000 0011 1011 0001

Table 1: A few examples of the mapping from glyph character to ASCII and Unicode. ASCII uses 7 bits to represent characters (with the eighth bit typically set to 0) and Unicode uses 16. The ASCII and Unicode mappings are compatible for the $2^7 = 128$ ASCII characters. The two glyphs on the bottom right of the table (the Greek letters, capital delta and lower case alpha) do not have encodings in ASCII.

Representing Numbers

8-digit Binary Number								
Value	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Position	7	6	5	4	3	2	1	0
Base 2	0	0	1	1	0	0	0	1
Base 10	0	0	32	16	0	0	0	1

Figure 1: The 8-digit binary number 00110001 is equivalent to the 2-digit decimal number 49. To see this, think of each position in the 8-digit number as representing a power of 2, from 2^0 to 2^7 . For a particular position, say the fifth position from the right, there appears a 1 in the number 001[1]0001 to indicate one 2^4 , or one 16. The decimal equivalent of the binary number is then the sum of the corresponding decimal values for each position that contains the digit 1.

Hexadecimal Numbers

- **Hexadecimal** (also **base 16**, or **hex**) is a [positional numeral system](#) with a [radix](#), or base, of 16.
- It uses sixteen distinct symbols, most often the symbols **0–9** to represent values zero to nine, and **A, B, C, D, E, F** (or alternatively **a–f**) to represent values ten to fifteen.
- We often start a hex number with a # symbol.
- For example, the hex number #2AF3 is equal, in decimal, to $(2 \times 16^3) + (10 \times 16^2) + (15 \times 16^1) + (3 \times 16^0) = 10995$, and the hex number #FF is equal, in decimal, to $(15 \times 16^1) + (15 \times 16^0) = 255$.

Representation of Colors

Colors: RGB (red, green & blue)

	(255, 0, 0)	#FF0000
	(255, 255, 0)	#FFFF00
	(100, 149, 237)	#6495ED
		#E41A1C99



You don't know what they are talking about ?? Try
`hist(0:1, col = c("#D0E8E6", "#F0F0D7"))`

Representation of Files

Plain text files

CSV: example.csv

```
1, 2, 30
12, 10, 11
```

HTML: example.html

```
<html>
<head>
<title> Title for Example page </title>
</head>
<body>
<h1> Example of HTML </h1>
<ul>
<li> A </li>
<li> <font color="red">B </font></li>
</ul>
</body>
</html>
```

Latex: example.tex

```
\documentclass[12pt]{article}
\begin{document}
\Large\textbf{Example of Latex}}
\begin{itemize} \item A \item B
\end{itemize}
\end{document}
```

Magic number

- postscript


```
%!PS-Adobe-2.0
%%Creator: dvips(k) 5.92b Copyright 2002 Radical Eye Software
%%Title: DataTypes.dvi
```
- PDF - portable document format


```
%PDF-1.4
3 0 obj <<
/Length 153
/Filter /FlateDecode
```
- PNG - portable network graphics (for images)


```
~IPNG~M
~Z
```

Representation of Data

HTML table, Excel Spreadsheet, plain
text

ManyEyes html

View as text

	Population Using Internet	Percent of Generation that goes online	Percent of the online population that watch video online
1 Millennials	35%	95%	80%
2 Gen X	21%	86%	66%
3 Younger Boomers	20%	81%	62%
4 Older Boomers	13%	76%	55%
5 Silent Generation	5%	58%	44%
6 G.I. Generation	3%	30%	20%
7 Total Population	79%	66%	

watch this add to topic center Visualize rate this

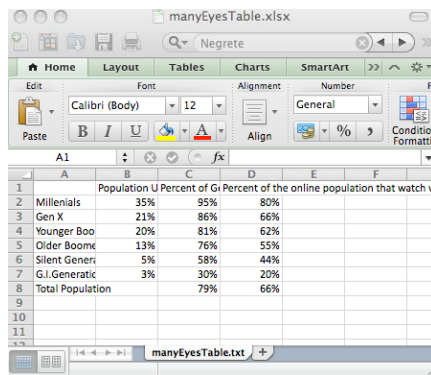
Versions (1)

<http://www-958.ibm.com/software/data/cognos/manyeyes/>

ManyEyes text

Population Using Internet			Percent of Generation
Millenials	35%	95%	80%
Gen X	21%	86%	66%
Younger Boomers	20%	81%	62%
Older Boomers	13%	76%	55%
Silent Generation	5%	58%	44%
G.I. Generation	3%	30%	20%
Total Population		79%	66%

ManyEyes xlsx



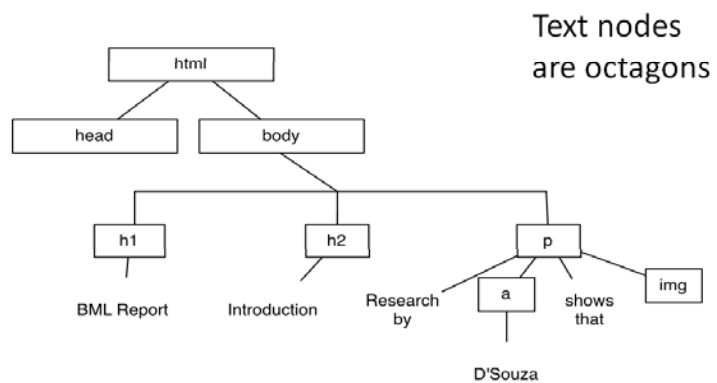
	Population U	Percent of G	Percent of the online population that watch vi
1 Millennials	35%	95%	80%
2 Gen X	21%	86%	66%
3 Younger Boo	20%	81%	62%
4 Older Boome	13%	76%	55%
5 Silent Generi	5%	58%	44%
6 G.I. Generatic	3%	30%	20%
7 Total Population		79%	66%

	txt	html	xlsx
browser	Render w/ no markup	Format according to markup	Open file in Excel
Excel	Display as Excel	Display as Excel	Display
TextEditor	Display ASCII characters	See markup as well as content	See nothing or jibberish

Hypertext Markup Language (HTML)

Tree

Tree Data Structure



Tree Hierarchy

- One root node
- Root node has child nodes and each of these can have child nodes and so on
- Any node must have one and only one parent

Examples of HTML

Table in HTML

A	B
1	25,000
7	100,000

`<table>`
`<tr>`
`<th>A</th> <th>B</th>`
`</tr>`
`<tr>`
`<td>1</td> <td>25,000</td>`
`</tr>`
`<tr>`
`<td>7</td>`
`<td>100,000</td> </tr>`
`</table>`

An HTML Table

- Tables are defined with the `<table>` tag.
- A table has rows marked up with the `<tr>` tag.
- Each row is divided into data cells with the `<td>` tag. (td stands for table data).
- A data cell can contain text, images, lists, paragraphs, forms, horizontal rules, tables, etc.
- Headings in a table are defined with the `<th>` tag.

Modified Table

A	B
1	25,000
7	100,000

`<table cellpadding="6" border="2">`
`<tr> <th>A</th> <th>B</th>`
`</tr>`
`<tr align="right">`
`<td>1</td><td>25,000</td>`
`</tr>`
`<tr align="right">`
`<td>7</td><td>100,000</td>`
`</tr>`
`</table>`

Unordered Lists

- Unordered lists have items marked with bullets.

Appears as:

```
<ul>
```

```
<li>Coffee</li>
```

```
<li>Milk</li>
```

```
</ul>
```

- Coffee
- Milk

- Paragraphs, line breaks, images, links, other lists, etc. can be placed in a list

Ordered Lists

- Ordered lists have items marked with numbers.

Appears as:

```
<ol>
```

```
<li>Coffee</li>
```

```
<li>Milk</li>
```

```
</ol>
```

1. Coffee

2. Milk

Paragraphs and Sections

```
<h1>
```

```
My BML Report
```

```
</h1>
```

```
<h2>Introduction</h2>
```

```
<p>
```

```
The BML model is a simple traffic model...
```

```
</p>
```

```
<p> We studied the BML model behavior for...
```

```
</p>
```

Appears as:

My BML Report

Introduction

The BML model is a simple traffic model...

We studied the BML model behavior for...

Images

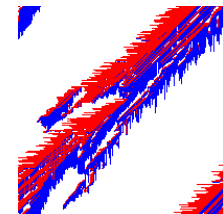
- The img tag is used to embed images in a Web page

Appears as:

```

```

- The src attribute give the file name for the image
- The width attribute is optional
- This tag is empty – the start and end tag are collapsed.



Links

`<a href="http://mae.ucdavis.edu/dsouza/">`
D'Souzza`</a>` discovered

Appears as: **Introduction**

[D'Souzza](http://mae.ucdavis.edu/dsouza/) discovered

`<a>` is an anchor tag

The content is the text that is “clickable”

The link can be to another place within the document

Element Syntax

- Each HTML element has an element name, e.g.
 - body : the main content of the page
 - h1 : largest header
 - p : paragraph
 - br : line break

Element Syntax

`<h1>` **This is a title** `</h1>`

Start tag Text Content End tag

- The end tag is a slash and the name surrounded by angle brackets `</h1>`
- Some HTML elements have no content `<br/>` is for a line break

Element Content

- Simple content is plain text:
`<h1> This is a title </h1>`
- Complex content includes other elements.
`<p>This paragraph includes <a href="http://...">a link</a> and sentences.</p>`
 How many child elements does this `<p>` node contain? 3

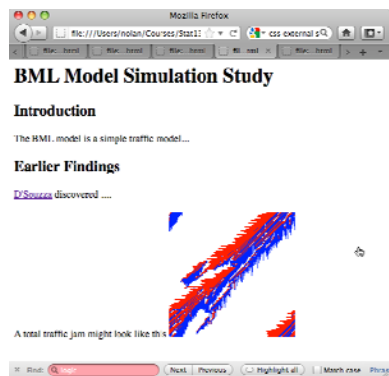
Attribute Syntax

- Attributes provide additional information to an HTML element.
- Attributes always come in name/value pairs like this: name="value"
- Attributes are always specified in the start tag of an HTML element.

Well-formed XHTML

- Well-formed HTML is called XHTML.
- Tag names follow strict rules for matching case
- Attribute values must be in quotes
- Elements must be properly nested (i.e. you can draw a tree with it)

A BML Report

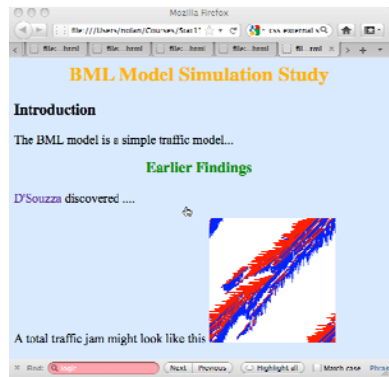


Raw HTML for the Stylized Report

```
<html>
<head></head>
<body>
<h1>BML Model Simulation Study</h1>
<h2>Introduction</h2>
<p> The BML model is a simple traffic model... </p>
<h2>Earlier Findings</h2>
<p>
<a href="http://mae.ucdavis.edu/dsouza/">D'Souza</a> discovered ....
</p>
<p>
A total traffic jam might look like this

</p>
</body>
</html>
```

A prettied up BML Report



Raw HTML for the Stylized Report

```
<html>
<head>
<link rel="stylesheet" type="text/css" href="bmlStyle.css" />
</head>
<body>
<h1>BML Model Simulation Study</h1>
<h2>Introduction</h2>
<p> The BML model is a simple traffic model... </p>
<h2 class="bml">Earlier Findings</h2>
<p>
<a href="http://mae.ucdavis.edu/dsouza/">D'Souza</a> discovered ....
</p>
<p>
A total traffic jam might look like this

</p>
</body>
</html>
```

Cascading Style Sheet (CSS)

```
body
{ background-color:#d0e4fe; }
h1
{ color:orange; text-align:center; }
h2.bml
{ color:green; text-align:center; }
p
{ font-family:"Times New Roman"; font-size:20px; }
```

CSS

selector {property: value; }

Selector may be:

- HTML tag name h1 {color: green;}
- attribute value for id #idXYZ {color:blue;}
- class .bml {font-size: 2em}